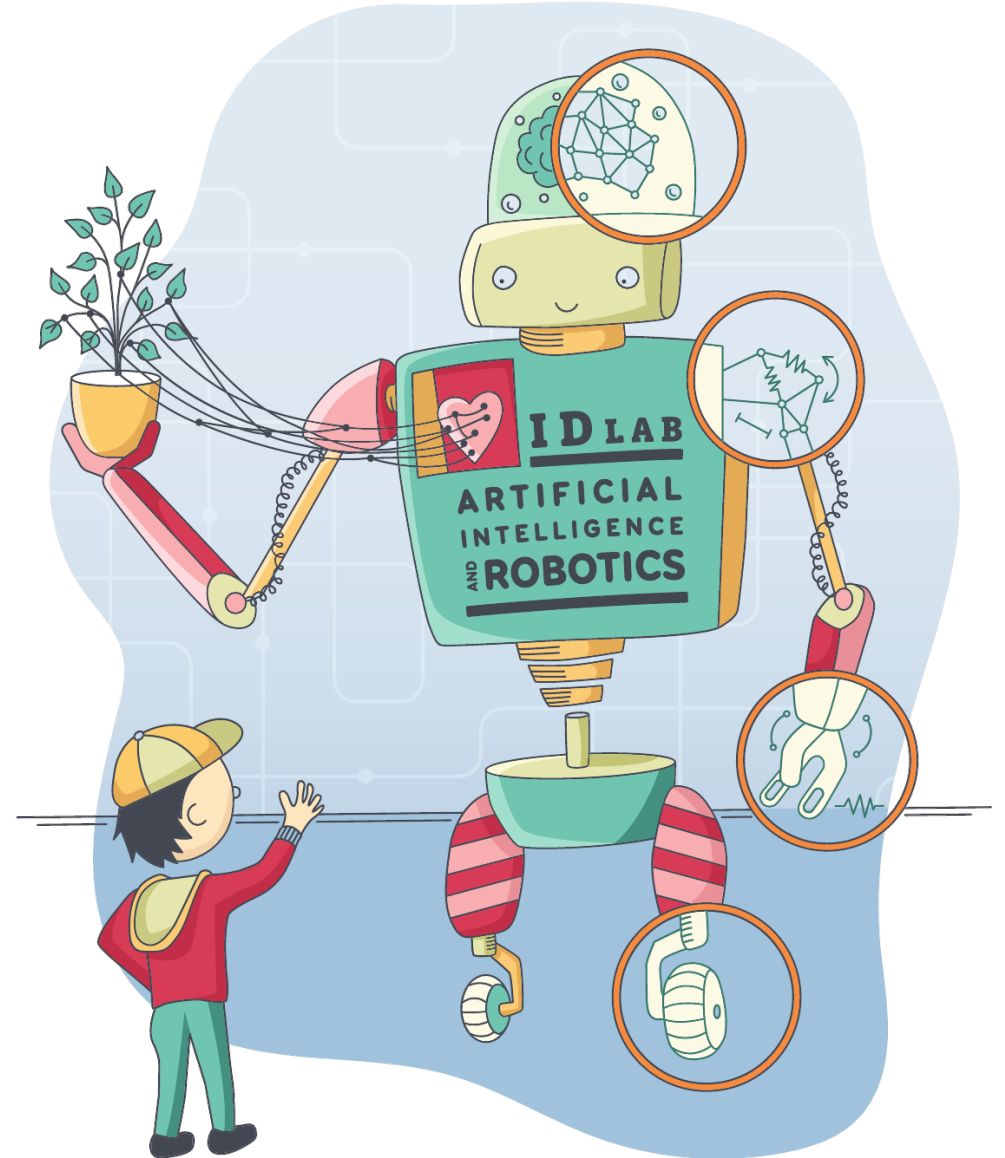
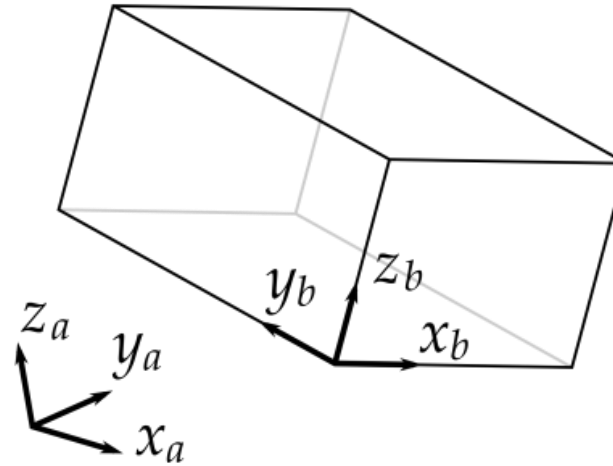


Kinematics for robotic manipulation

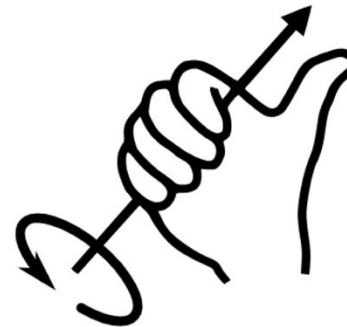
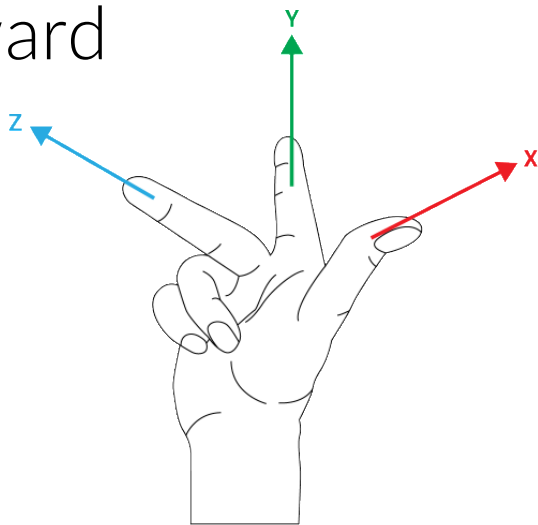


Previously: spatial algebra

- Pose of object -> frame
- Homogeneous transformation matrix
- Colours **R**, **G**, **B** are also used for x, y, z respectively
- z axis is upward
- $x \times y = z$

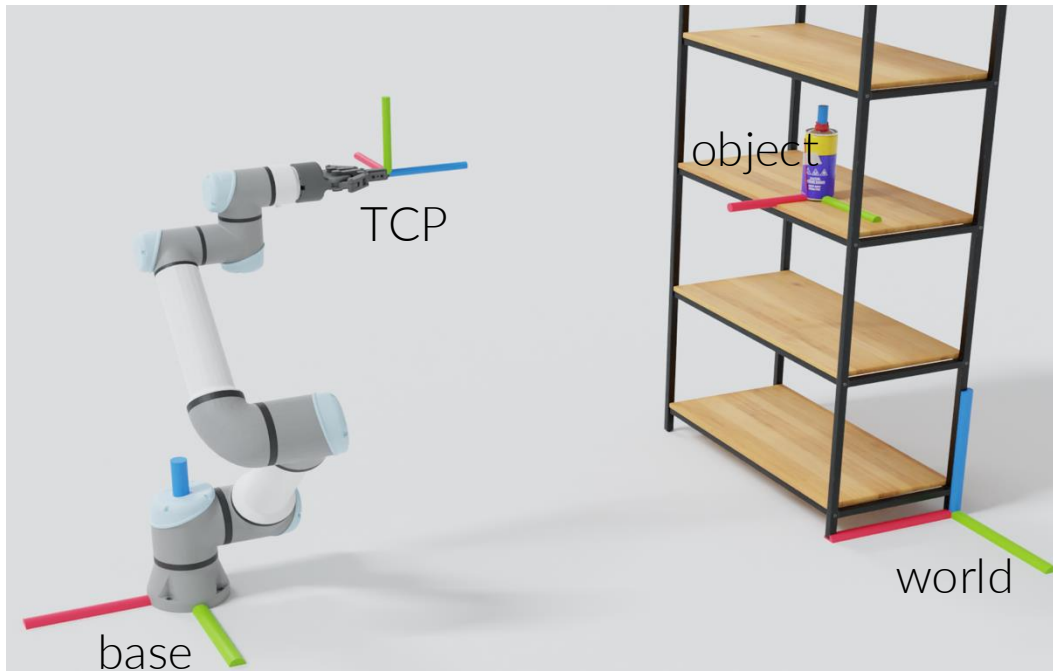


$$X = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_1 \\ r_{21} & r_{22} & r_{23} & p_2 \\ r_{31} & r_{32} & r_{33} & p_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Quiz

Given the following scenario

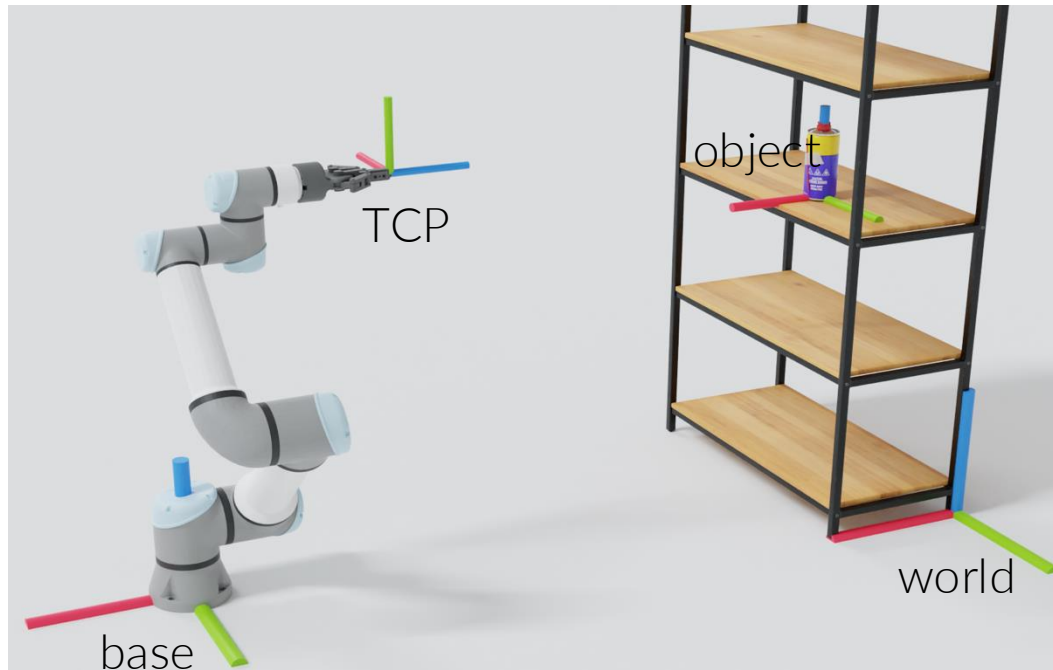


What is a possible value for ${}^{TCP}p_{object}$?

- a. $[5., -0.1, 0.1]$
- b. $[-0.1, -0.1, 5.]$
- c. $[0.1, 0.1, -5.]$
- d. $[-5., 0.1, -0.1]$

Quiz

Given the following scenario



Which frame do I send to the robot when I want to grasp the object?

a.
$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

b.
$$\begin{bmatrix} 0 & 0 & -1 & -7 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

c.
$$\begin{bmatrix} 0 & -1 & 0 & -7 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

d.
$$\begin{bmatrix} 0 & 0 & -1 & 1 \\ -1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

e.
$$\begin{bmatrix} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Forward kinematics

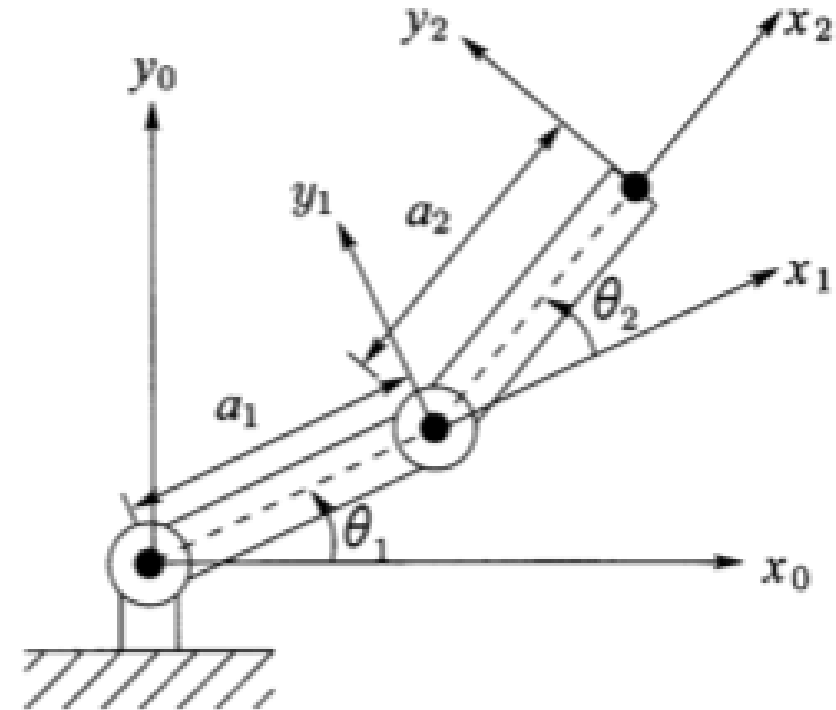
Forward kinematics: planar robot

Given the joint positions q what will be the end-effector's pose?

Tip: in a right-angled triangle

$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$$



Forward kinematics: planar robot

Given the joint positions q what will be the end-effector's pose?

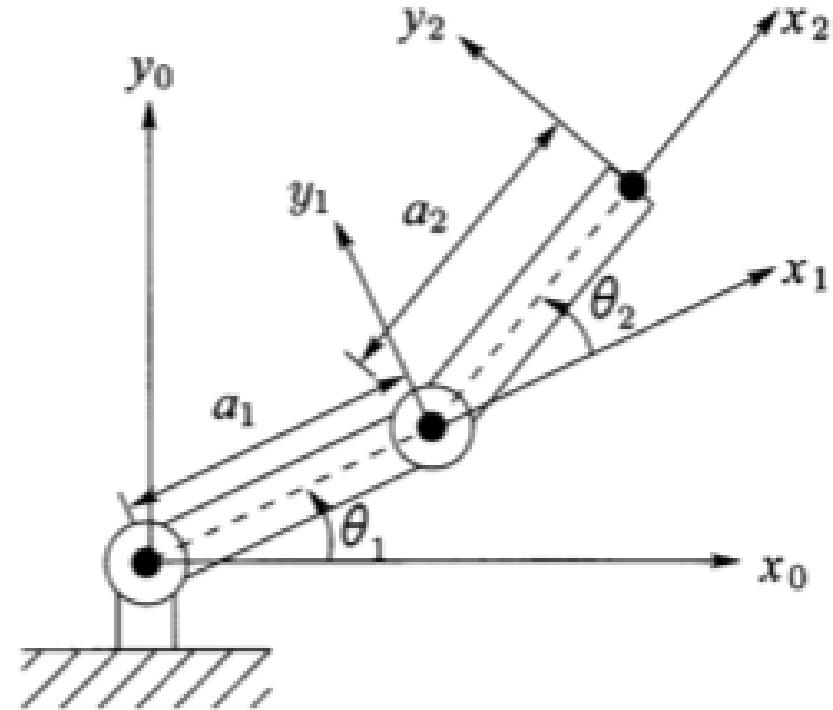
$$x_2 = a_1 \cos(q_1) + a_2 \cos(q_1 + q_2)$$

$$y_2 = a_1 \sin(q_1) + a_2 \sin(q_1 + q_2)$$

In short:

$$x_2 = a_1 c_1 + a_2 c_{12}$$

$$y_2 = a_1 s_1 + a_2 s_{12}$$



Forward kinematics: planar robot

Given the joint positions q what will be the end-effector's pose?

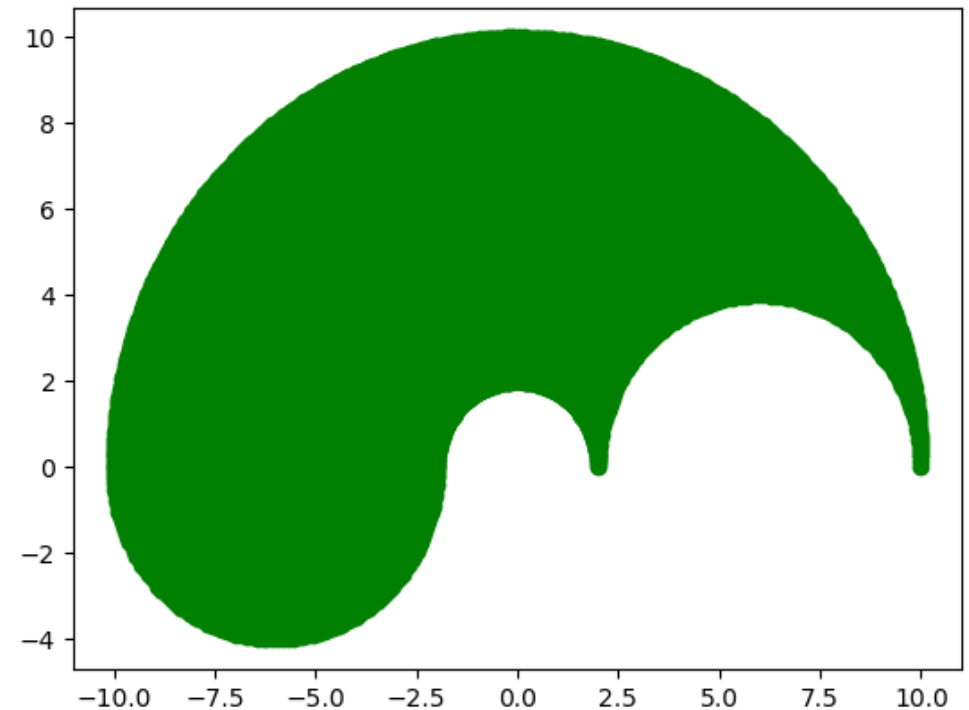
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In short:

$$x_2 = a_1 c_1 + a_2 c_{12}$$

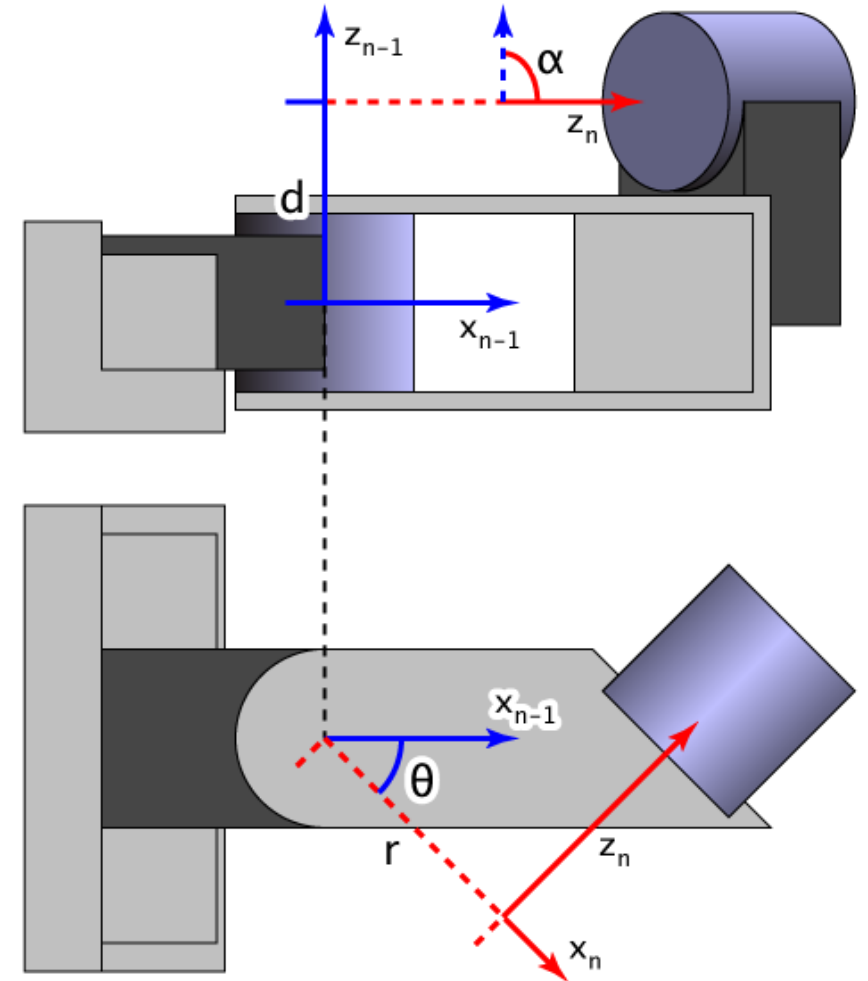
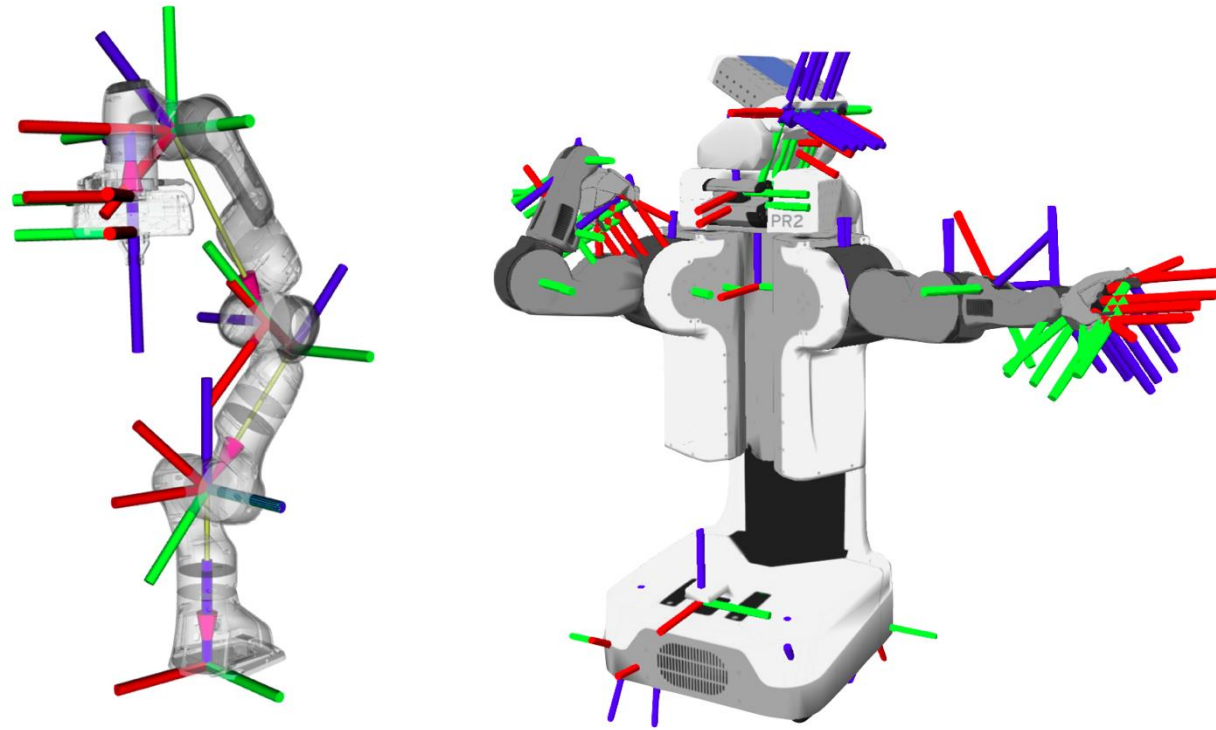
$$y_2 = a_1 s_1 + a_2 s_{12}$$



Workspace for $a_1 = 6$, $a_2 = 4$

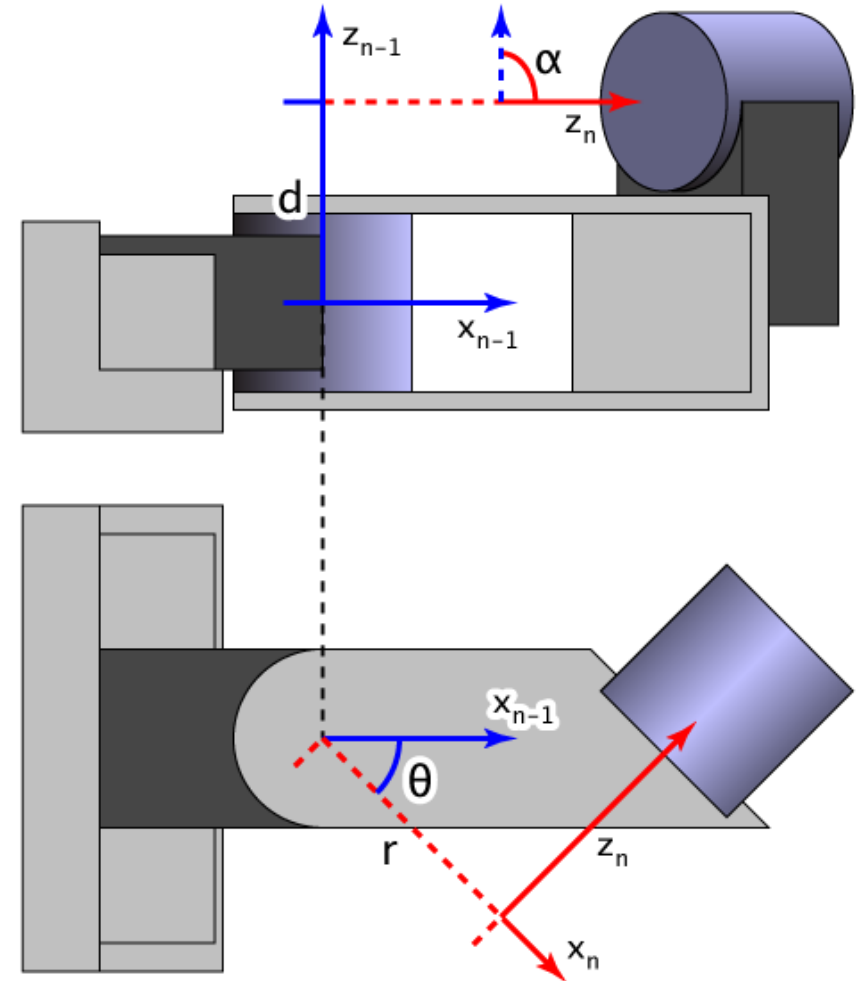
Kinematic chain

A robot arm consists of multiple rigid bodies connected with each other via a joint

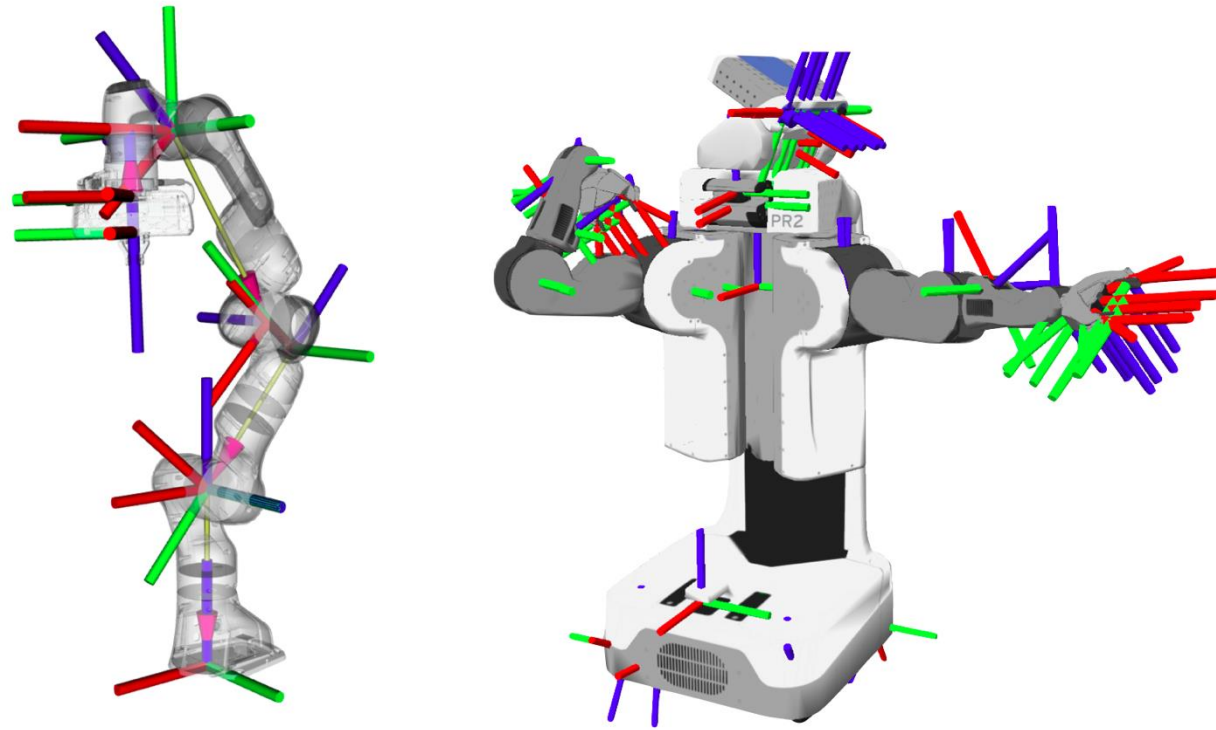


A robot arm consists of multiple rigid bodies connected with each other via a joint

$${}^{n-1}X^n = \begin{bmatrix} {}^{n-1}R^n & {}^{n-1}p^n \\ 0 & 1 \end{bmatrix}$$



A robot arm consists of multiple rigid bodies connected with each other via a joint



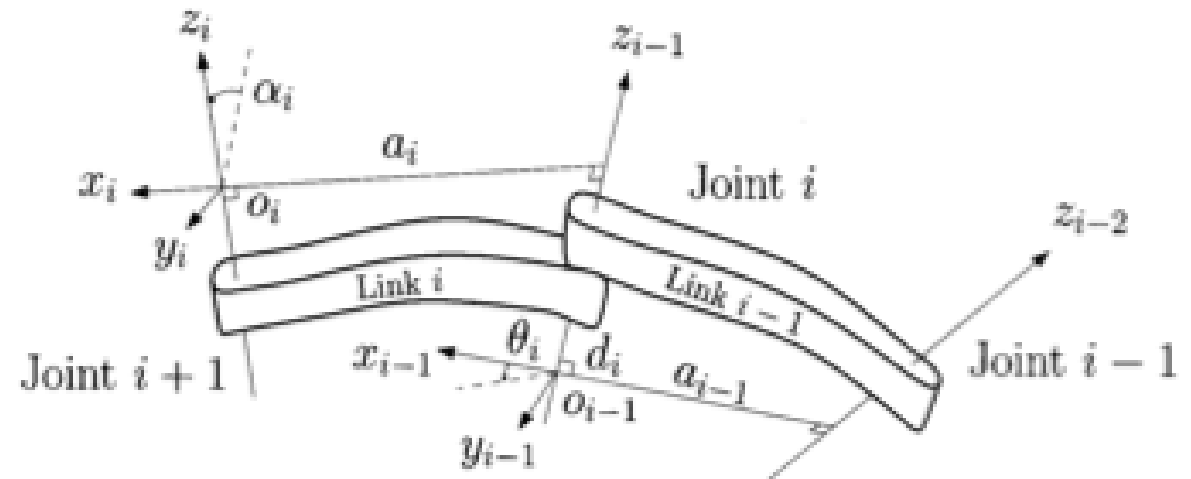
$${}^0X^N = {}^0X^1 {}^1X^2 \dots {}^{N-1}X^N$$

Note that this determines the forward kinematics

Denavit-Hartenberg convention

Defines a set of rules for assigning the link frames so that the kinematics can be described with four parameters (instead of 6 parameters)

- Axiom 1: x_{i+1} is perpendicular to z_i
- Axiom 2: x_{i+1} intersects z_i

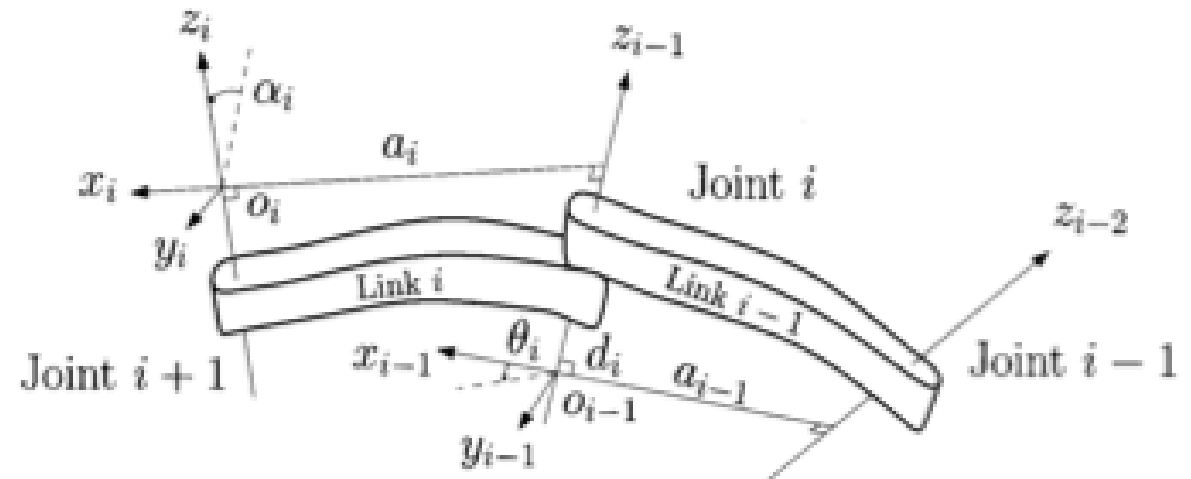


Denavit-Hartenberg convention

Defines a set of rules for assigning the link frames so that the kinematics can be described with four parameters

- **Link length** a_i : defines from z_{i-1} and z_i measured along x_i
- **Link twist** α_i : is the angle from z_{i-1} to z_i over x_i
- **Link offset** d_i : the offset of the link measured from x_{i-1} to x_i along z_{i-1}
- **Joint angle** θ_i : rotation of the link from x_{i-1} to x_i measured over z_{i-1}

Note: there exist multiple conventions, we use the *standard DH convention*



Denavit-Hartenberg convention

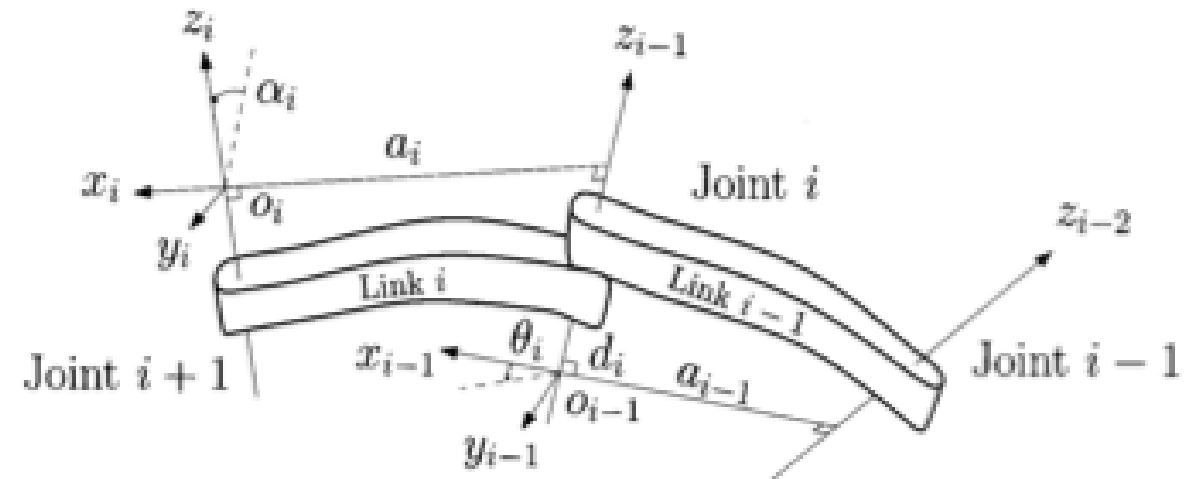
$${}^{i-1}X^i = \text{Rot}(\hat{z}, \theta_i) \text{Trans}(\hat{z}, d_i) \text{Trans}(\hat{x}, a_i) \text{Rot}(\hat{x}, \alpha_i)$$

$$\text{Rot}(\hat{x}, \alpha_i) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha_i & -\sin \alpha_i & 0 \\ 0 & \sin \alpha_i & \cos \alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$\text{Trans}(\hat{x}, a_i) = \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$\text{Trans}(\hat{z}, d_i) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

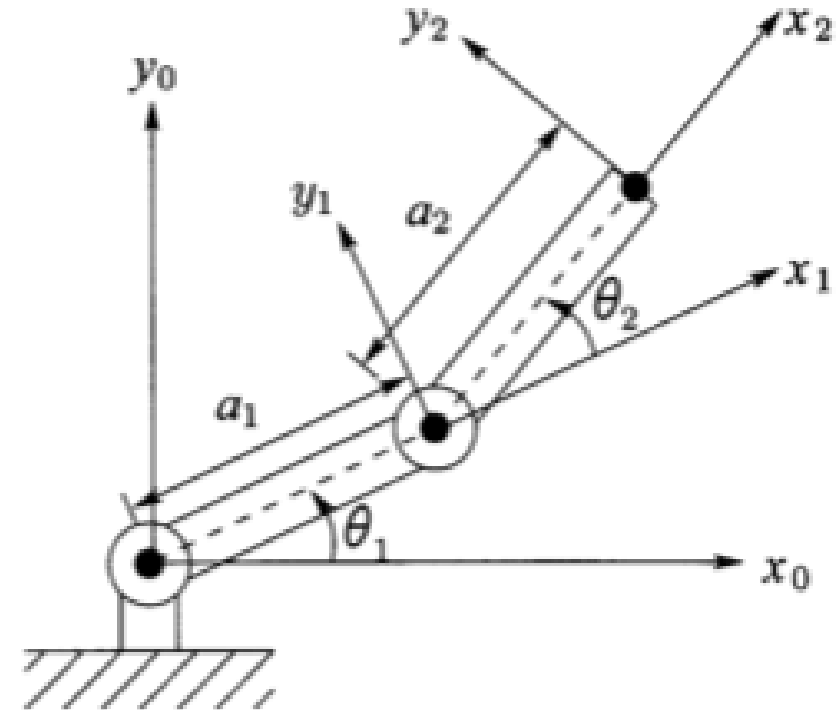
$$\text{Rot}(\hat{z}, \theta_i) = \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & 0 \\ \sin \theta_i & \cos \theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$



Denavit-Hartenberg: exercise

Make the D-H table for this 2-segmented planar robot

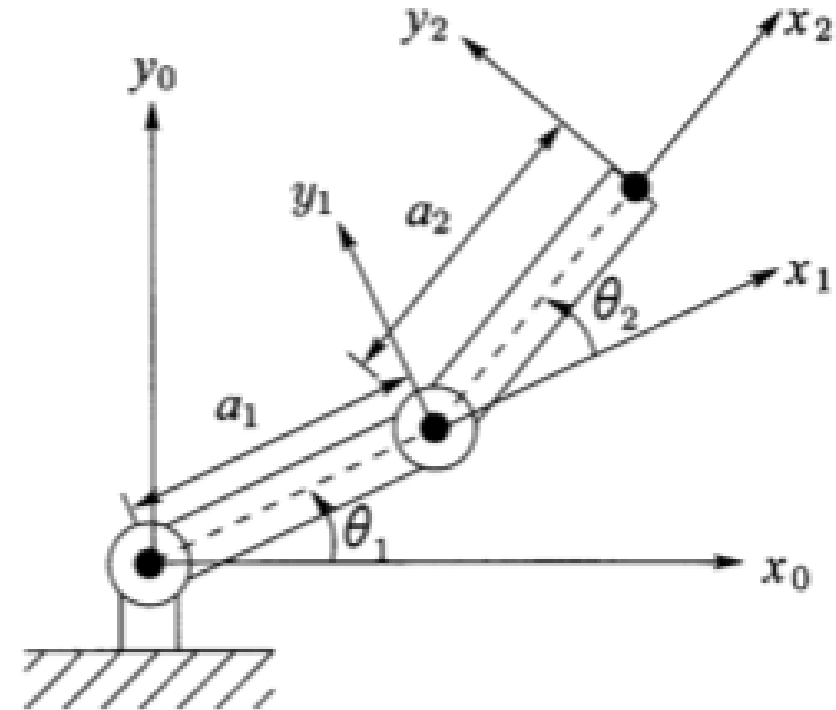
Joint	θ_i	d_i	a_i	α_i
1				
2				



Denavit-Hartenberg: exercise

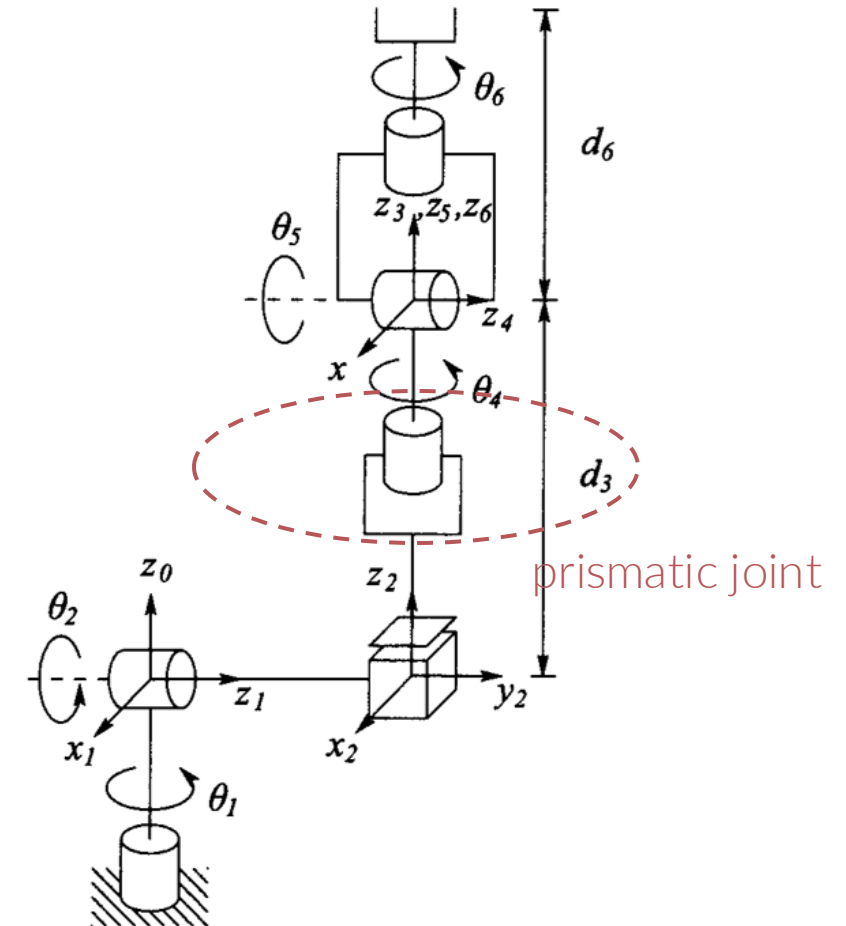
Make the D-H table for this 2-segmented planar robot

Joint	θ_i	d_i	a_i	α_i
1	θ_1^*	0	a_1	0
2	θ_2^*	0	a_2	0



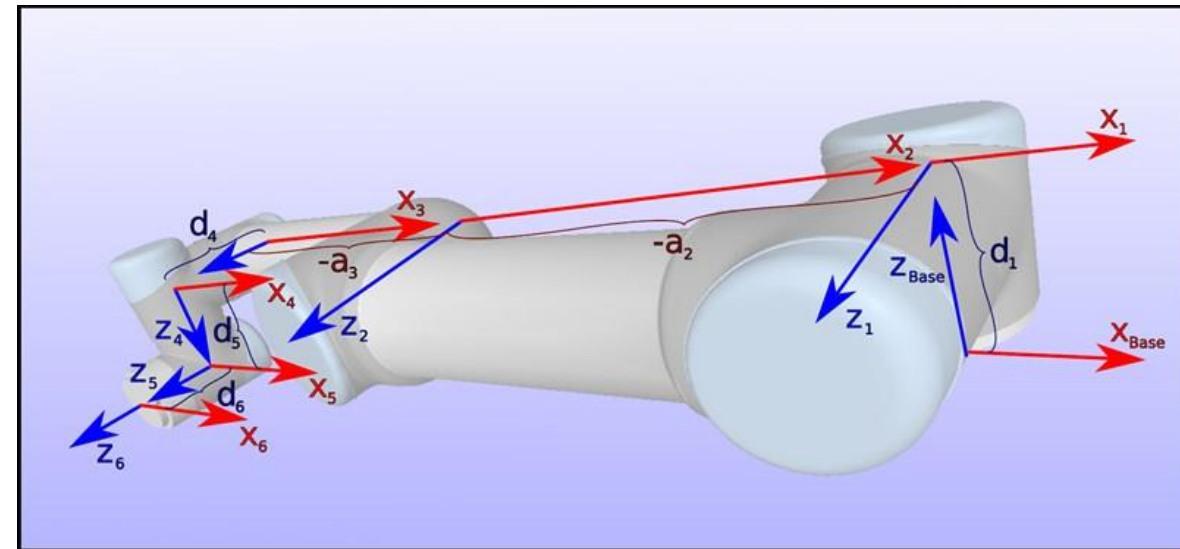
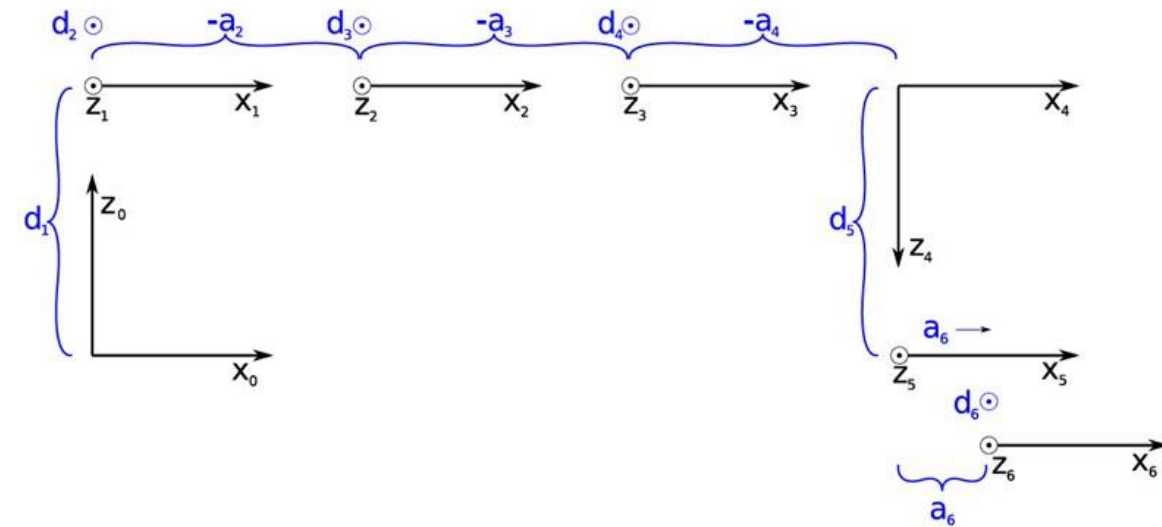
Denavit-Hartenberg: Stanford manipulator

Joint	θ_i	d_i	a_i	α_i
1	θ_1^*	0	0	$-\pi/2$
2	θ_2^*	d_2	0	$\pi/2$
3	0	d_3^*	0	0
4	θ_4^*	0	0	$-\pi/2$
5	θ_5^*	0	0	$\pi/2$
6	θ_6^*	d_6	0	0



Denavit-Hartenberg: UR3e robot

Joint	d_i [m]	θ_i [rad]	a_i [m]	α_i [rad]
1	$d_1=0.15185$	θ_1^*	0	$\pi/2$
2	0	θ_2^*	-0.24355	0
3	0	θ_3^*	-0.2132	0
4	$d_4=0.13105$	θ_4^*	0	$\pi/2$
5	$d_5=0.08535$	θ_5^*	0	$-\pi/2$
6	$d_6=0.0921$	θ_6^*	0	0



Forward kinematics (bis)

Forward kinematics: planar robot

Given the joint positions q what will be the end-effector's pose?

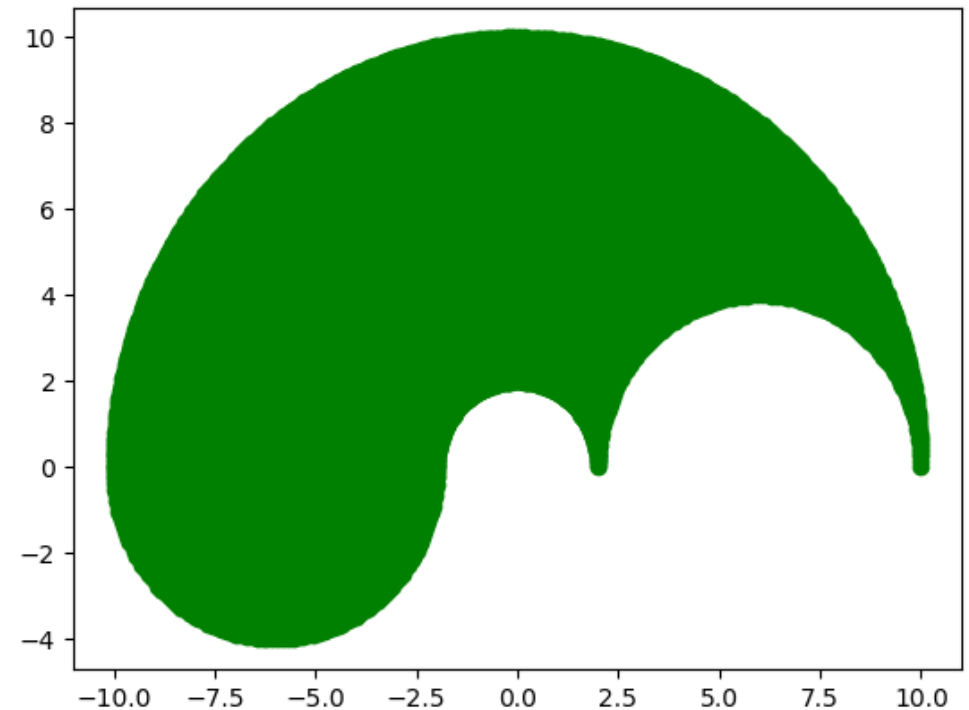
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$$y_2 = a_1 \sin(q_1) + a_2 \sin(q_1 + q_2)$$

In short:

$$x_2 = a_1 c_1 + a_2 c_{12}$$

$$y_2 = a_1 s_1 + a_2 s_{12}$$



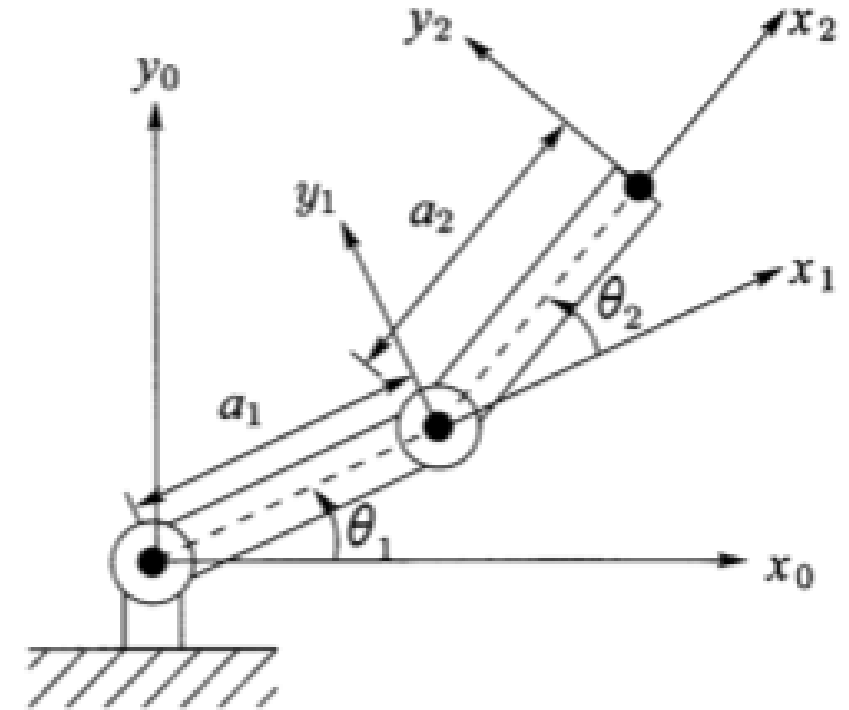
Workspace for $a_1 = 6$, $a_2 = 4$

Forward kinematics: planar robot (bis)

Given the joint positions q what will be the end-effector's pose?

Joint	θ_i	d_i	a_i	α_i
1	θ_1^*	0	a_1	0
2	θ_2^*	0	a_2	0

$${}^0X^2 = ?$$

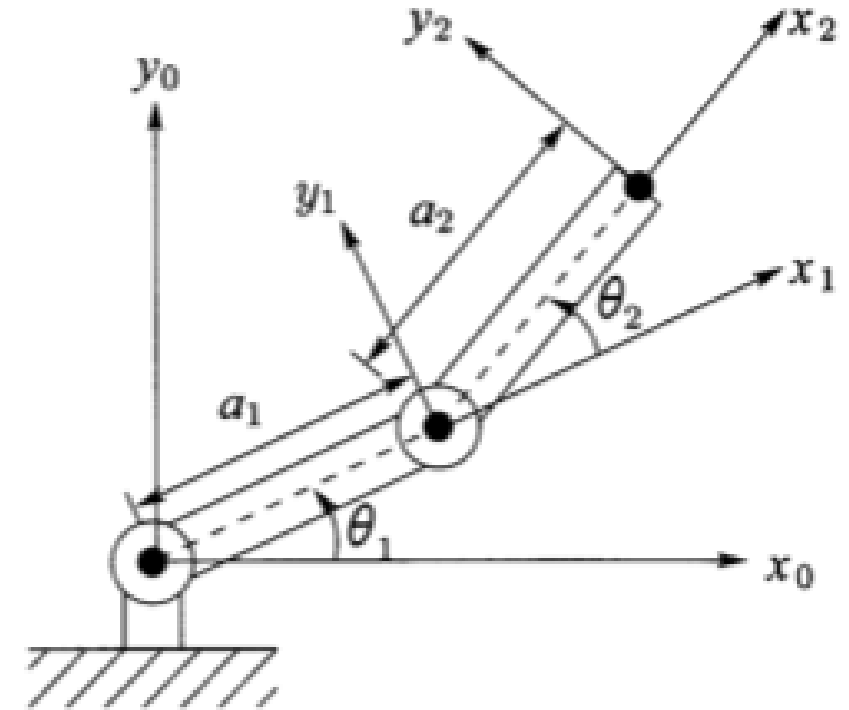


Forward kinematics: planar robot (bis)

Given the joint positions q what will be the end-effector's pose?

Joint	θ_i	d_i	a_i	α_i
1	θ_1^*	0	a_1	0
2	θ_2^*	0	a_2	0

$${}^0X^2 = \begin{bmatrix} c_{12} & -s_{12} & 0 & a_1c_1 + a_2c_{12} \\ s_{12} & c_{12} & 0 & a_1s_1 + a_2s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Inverse kinematics

Inverse kinematics: planar robot

Given the end-effector's pose what should the joint positions q be?

$$x_2 = L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2)$$

$$y_2 = L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2)$$

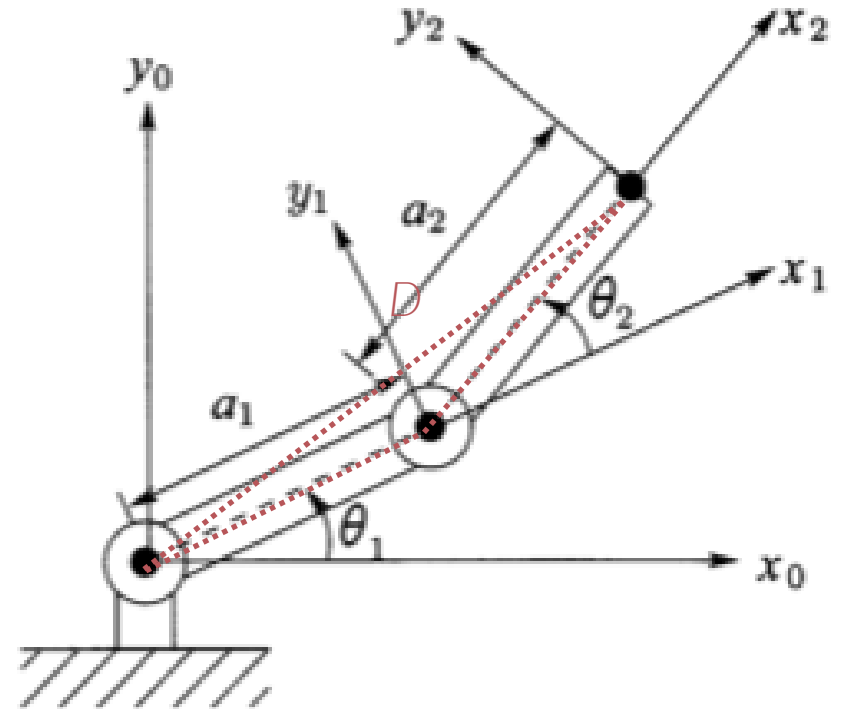
Write θ_1, θ_2 in function of x_2, y_2

Tip: cosine rule in triangle $a_1 a_2 D$

$$D^2 = L_1^2 + L_2^2 - 2L_1 L_2 \cos(\phi)$$

$$\sin(\theta_1 + \theta_2) = \sin(\theta_1) \cos(\theta_2) + \cos(\theta_1) \sin(\theta_2)$$

$$\cos(\theta_1 + \theta_2) = \cos(\theta_1) \cos(\theta_2) - \sin(\theta_1) \sin(\theta_2)$$



Inverse kinematics: planar robot

Given the **end-effector's pose** what should the **joint positions q** be?

$$D^2 = x_2^2 + y_2^2 = L_1^2 + L_2^2 - 2L_1L_2 \cos(\phi)$$

$$\phi = \pi - \theta_2$$

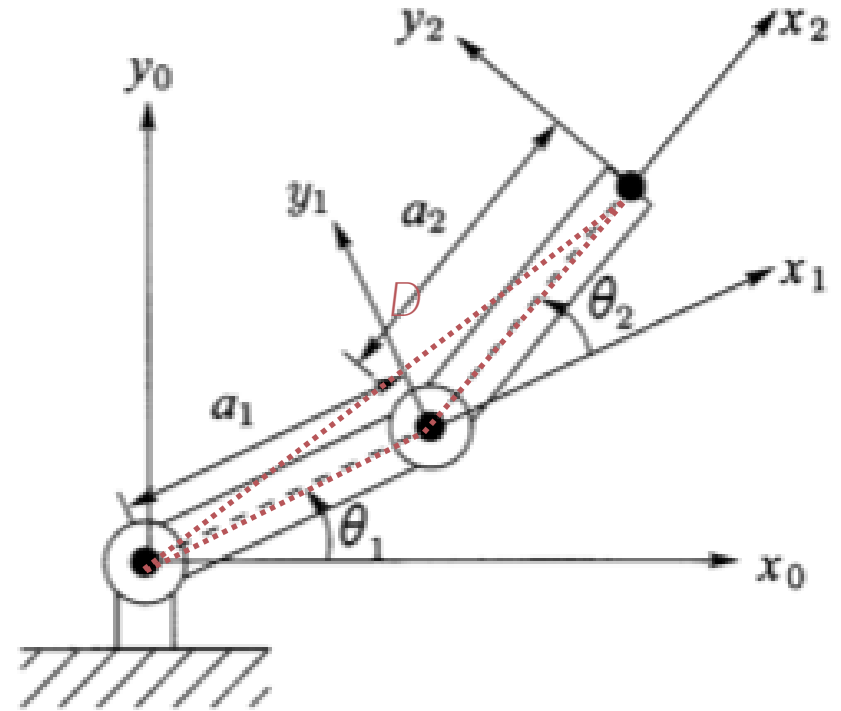
$$x_2^2 + y_2^2 = L_1^2 + L_2^2 - 2L_1L_2 \cos(\pi - \theta_2)$$

$$x_2^2 + y_2^2 = L_1^2 + L_2^2 + 2L_1L_2 \cos(\theta_2)$$

$$x_2^2 + y_2^2 = L_1^2 + L_2^2 + 2L_1L_2 c_2$$

$$c_2 = \frac{x_2^2 + y_2^2 - L_1^2 - L_2^2}{2L_1L_2}$$

$$\theta_2 = \pm \cos^{-1}\left(\frac{x_2^2 + y_2^2 - L_1^2 - L_2^2}{2L_1L_2}\right)$$



Inverse kinematics: planar robot

Given the end-effector's pose what should the joint positions q be?

$$c_2 = \frac{x_2^2 + y_2^2 - L_1^2 + L_2^2}{2L_1L_2}$$

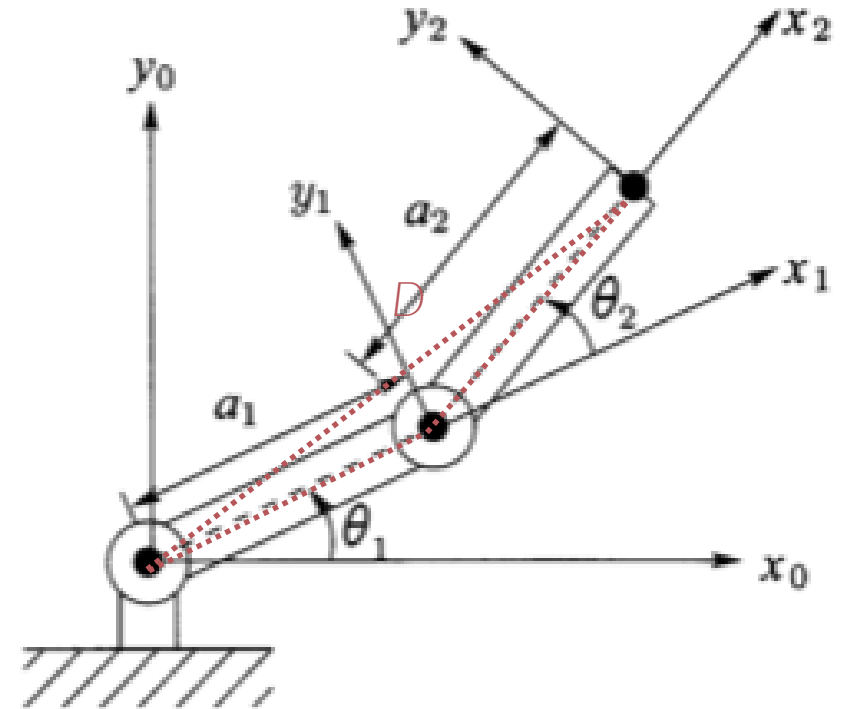
$$\theta_2 = \pm \cos^{-1}\left(\frac{x_2^2 + y_2^2 - L_1^2 + L_2^2}{2L_1L_2}\right)$$

$$\begin{cases} x_2c_1 = L_1c_1^2 + L_2c_1^2c_2 - L_2c_1s_1s_2 \\ y_2s_1 = L_1s_1^2 + L_2s_1^2c_2 + L_2s_1c_1s_2 \end{cases}$$

$$\begin{cases} x_2c_1 + y_2s_1 = L_1 + L_2c_2 \\ -x_2s_1 + y_2c_1 = L_2s_2 \end{cases}$$

$$c_1(x_2^2 + y_2^2) = x_2(L_1 + L_2c_2) + y_2L_2s_2$$

$$\theta_1 = \pm \cos^{-1}\left(\frac{x_2(L_1 + L_2c_2) + y_2L_2s_2}{x_2^2 + y_2^2}\right)$$



Inverse kinematics: analytical method

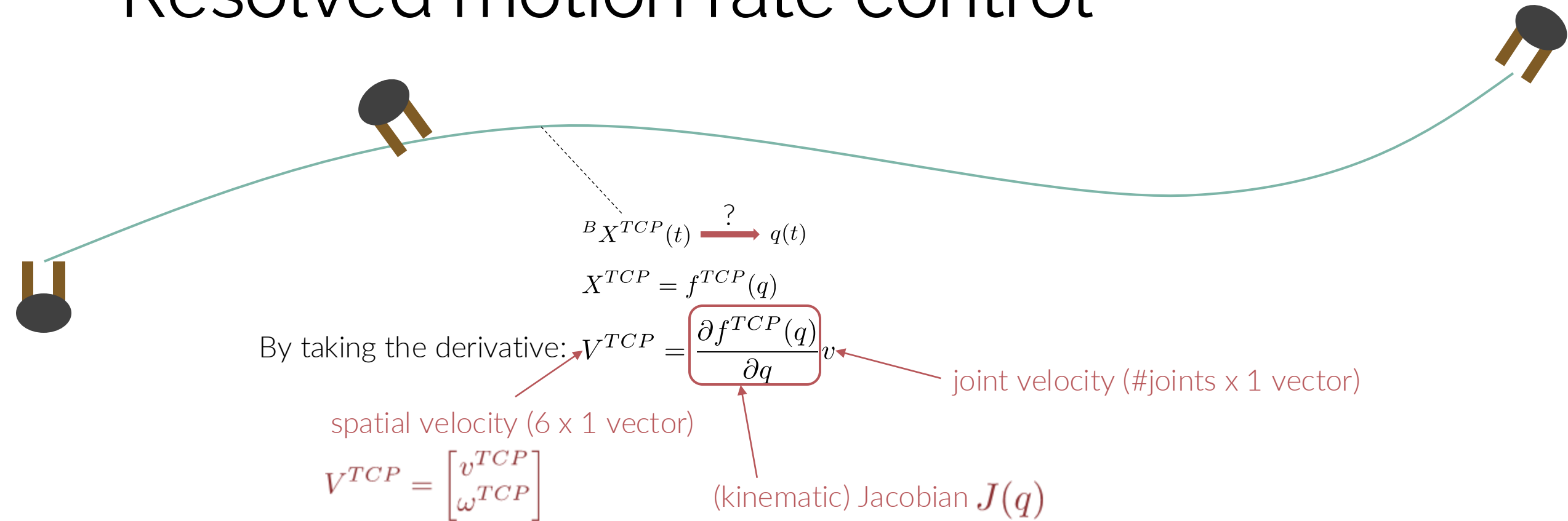
Given the **end-effector's pose** what should the **joint positions q** be?

- Analytical method
- Not always easy to find a closed form solution
- If you find a solution, it's a very fast method
- Sometimes multiple solutions as well
- Doesn't scale up easily, curious for a closed form solution for the UR? -> see optional reading material

Inverse kinematics (bis)

Differential inverse kinematics

Resolved motion rate control



The Jacobian matrix

$$J(q) = \begin{bmatrix} \frac{\partial x_1}{\partial \theta_1} & \cdots & \frac{\partial x_1}{\partial \theta_N} \\ \cdots & \cdots & \cdots \\ \frac{\partial x_M}{\partial \theta_1} & \cdots & \frac{\partial x_M}{\partial \theta_N} \end{bmatrix} \begin{matrix} \updownarrow \\ 6 \end{matrix} = \begin{bmatrix} J_v \\ J_\omega \end{bmatrix}$$

$\longleftrightarrow$
joints

The Jacobian matrix: angular velocity

$$J(q) = \begin{bmatrix} J_v \\ J_\omega \end{bmatrix}$$

$$J_\omega = \begin{bmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \rho_1 & \rho_2 & \dots & \rho_N \end{bmatrix}$$

$$\rho_i = \begin{cases} 0 & \text{for prismatic joint,} \\ 1 & \text{for revolute joint over } z_i \end{cases}$$

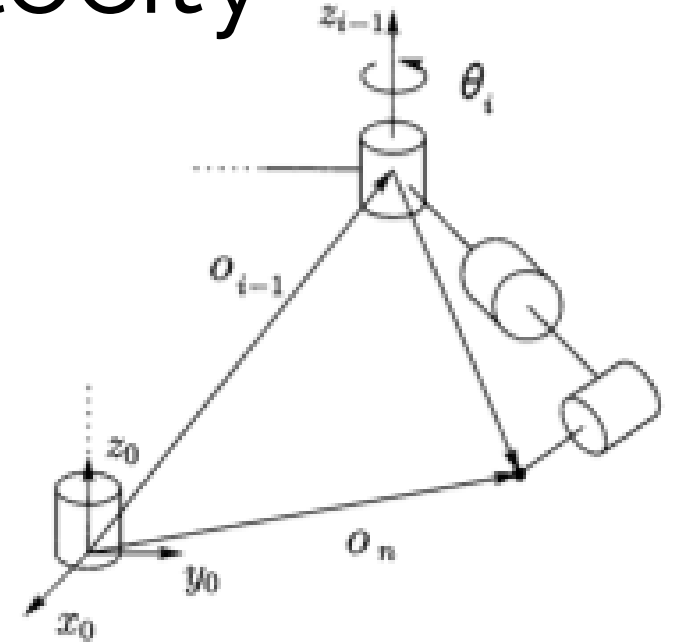


The Jacobian matrix: linear velocity

$$J(q) = \begin{bmatrix} J_v \\ J_\omega \end{bmatrix}$$

$$J_v = [J_{v_1} \quad J_{v_2} \quad \dots \quad J_{v_N}]$$

$$J_{v_i} = \begin{cases} z_i & \text{for prismatic joint,} \\ z_i \times (o_n - o_{i-1}) & \text{for revolute joint} \end{cases}$$



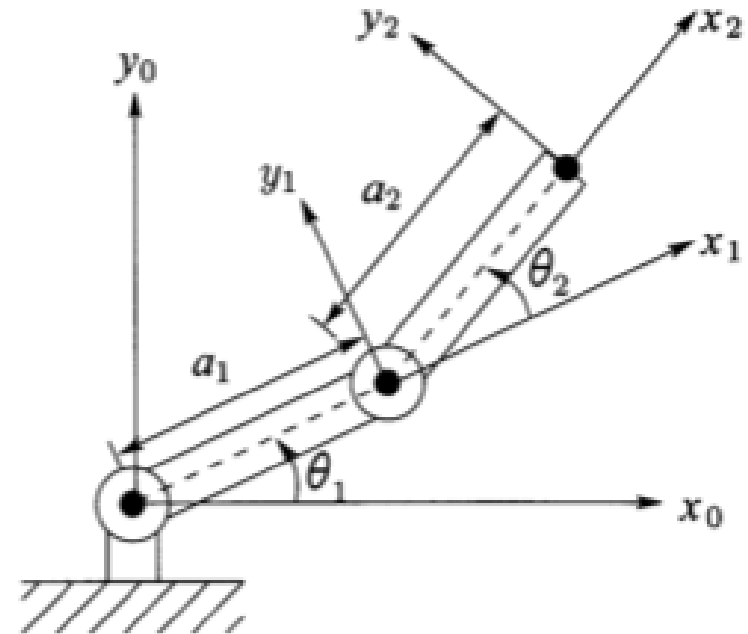
The Jacobian matrix: example

Joint	θ_i	d_i	a_i	α_i
1	θ_1^*	0	a_1	0
2	θ_2^*	0	a_2	0

$${}^0X^2 = \begin{bmatrix} c_{12} & -s_{12} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{12} & c_{12} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$J(q) = \begin{bmatrix} J_v \\ J_\omega \end{bmatrix}$$

Dimension?

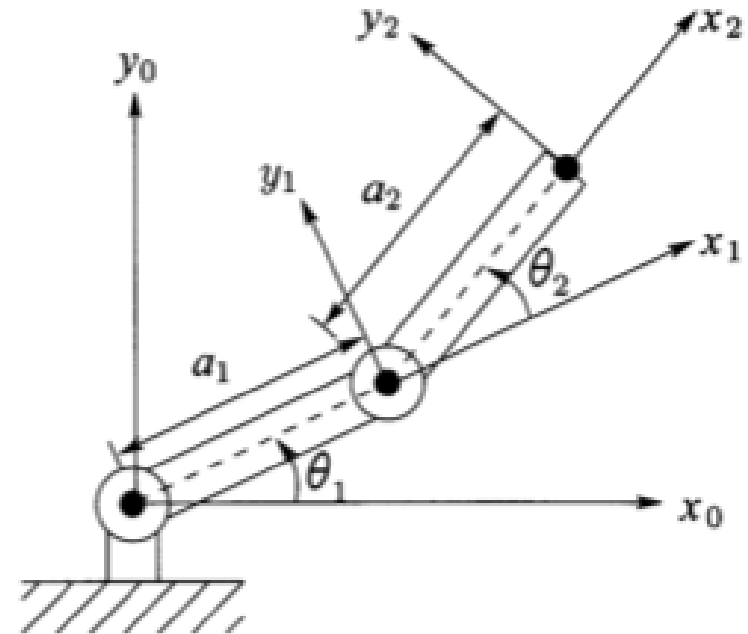


The Jacobian matrix: example

Joint	θ_i	d_i	a_i	α_i
1	θ_1^*	0	a_1	0
2	θ_2^*	0	a_2	0

$${}^0X^2 = \begin{bmatrix} c_{12} & -s_{12} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{12} & c_{12} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$J(q) = \begin{bmatrix} J_v \\ J_\omega \end{bmatrix} \quad J_\omega = ?$$

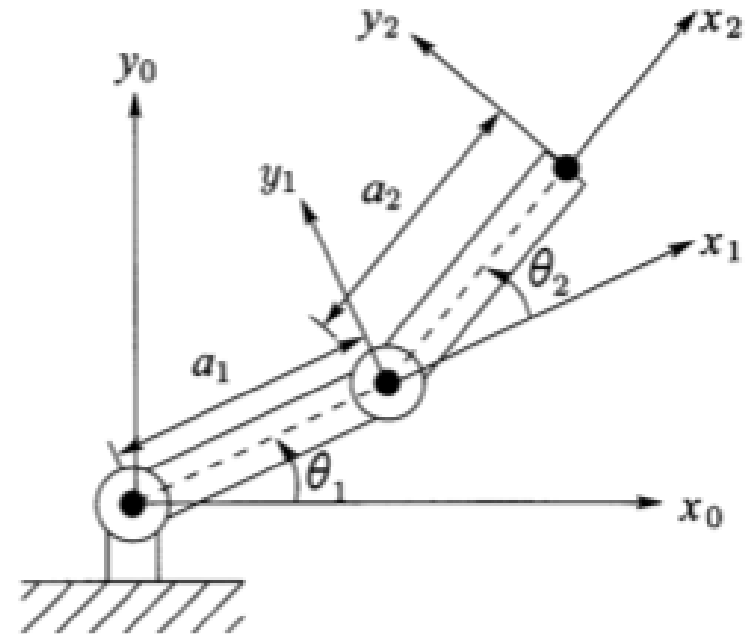


The Jacobian matrix: example

Joint	θ_i	d_i	a_i	α_i
1	θ_1^*	0	a_1	0
2	θ_2^*	0	a_2	0

$${}^0X^2 = \begin{bmatrix} c_{12} & -s_{12} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{12} & c_{12} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$J(q) = \begin{bmatrix} J_v \\ J_\omega \end{bmatrix} \quad J_\omega = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$$

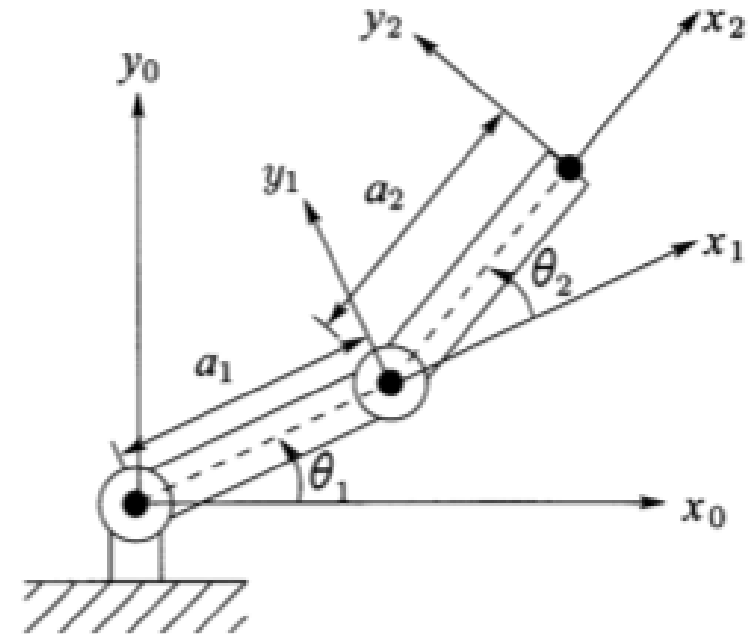


The Jacobian matrix: example

Joint	θ_i	d_i	a_i	α_i
1	θ_1^*	0	a_1	0
2	θ_2^*	0	a_2	0

$${}^0X^2 = \begin{bmatrix} c_{12} & -s_{12} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{12} & c_{12} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$J(q) = \begin{bmatrix} J_v \\ J_\omega \end{bmatrix} \quad J_v = ?$$



The Jacobian matrix: example

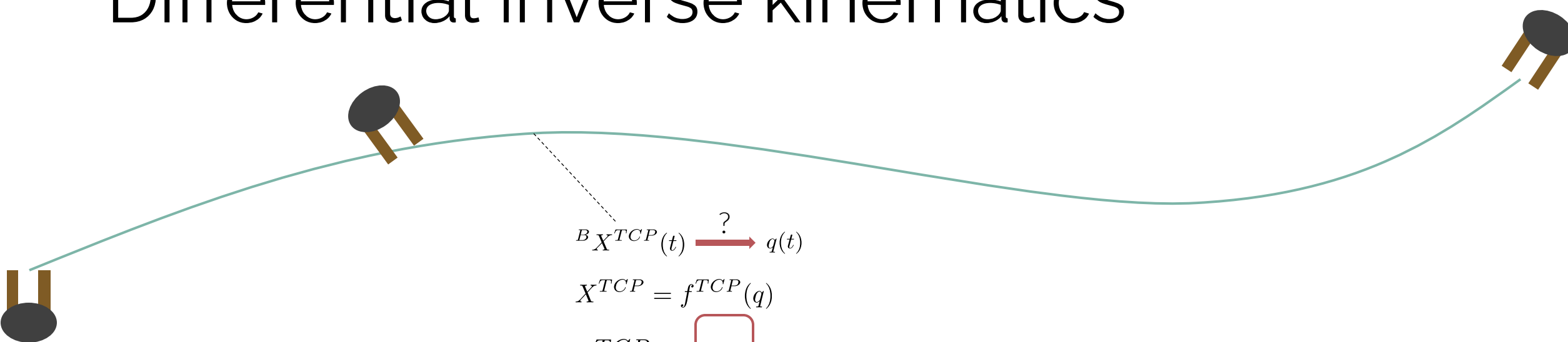
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1	θ_1^*	0	a_1	0
2	θ_2^*	0	a_2	0

$${}^0X^2 = \begin{bmatrix} c_{12} & -s_{12} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{12} & c_{12} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$J(q) = \begin{bmatrix} J_v \\ J_\omega \end{bmatrix} \quad J_v = \begin{bmatrix} -a_1 s_1 - a_2 s_{12} & -a_2 s_{12} \\ a_1 c_1 + a_2 c_{12} & a_2 c_{12} \\ 0 & 0 \end{bmatrix}$$

Note: you could have used the formula from before or by taking the derivatives

Differential inverse kinematics



$${}^B X^{TCP}(t) \xrightarrow{?} q(t)$$

$$X^{TCP} = f^{TCP}(q)$$

By taking the derivative: $V^{TCP} = J(q)v$

Configurations (determined by q) for which the rank of the Jacobian is less than its maximum value are called singularities

In the case of a square Jacobian matrix: this is the case when the determinant equals zero



Singularities

- Singularities represent configurations from which certain directions of motion are unattainable
- At singularities, bounded end-effector velocities may result in unbounded joint velocities
- At singularities, bounded joint torques may result in unbounded end-effector forces and torques

Singularities in practice



Differential inverse kinematics: an algorithm

1. Calculate the error e between the actual and desired pose;
2. Compute the Jacobian;
3. Invert* the Jacobian;
4. Calculate $\Delta\theta = J^*e$
5. Move $\theta = \theta + \alpha\Delta\theta$
6. Repeat 1-5 until the error e gets small

Multiple options:

- transpose
- pseudo inverse
- ...

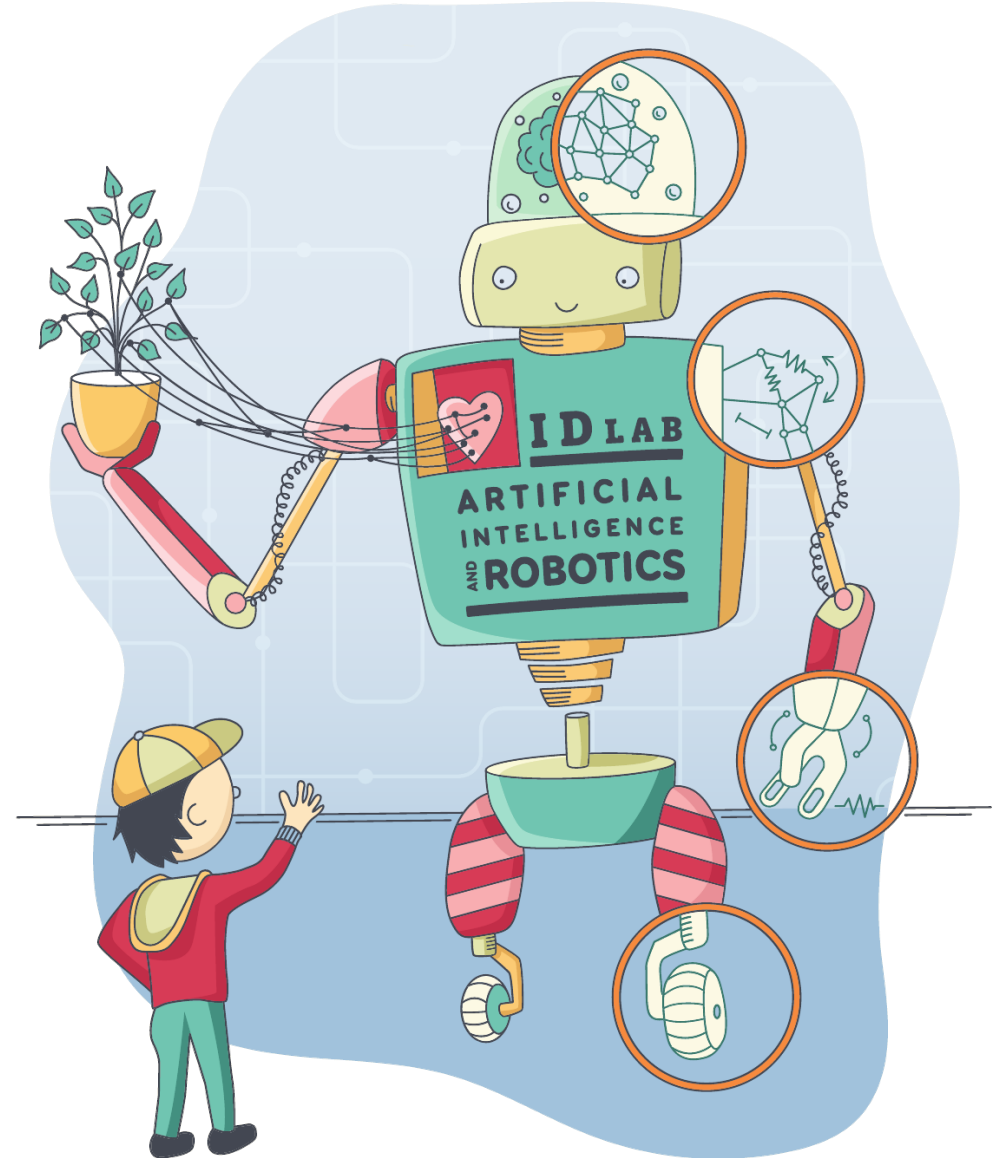
Differential inverse kinematics: an optimisation problem

From linear algebra we know that the pseudo-inverse is the optimal solution to a least-squares optimisation problem:

$$\min_v ||J(q)v - V||^2$$

This opens possibilities for adding additional constraints (e.g., on the velocities) or to use advanced solvers

Next week we
will bring these
concepts into
practice 😊 Don't
forget the
homework!



Homework

Given the DH table of the UR3e robot on the slide of the course material

1. Determine ${}^0X^N = {}^0X^1 {}^1X^2 \dots {}^{N-1}X^N$ (N=6)
2. Write a small python program with which you visualise the workspace of the UR3e robot by sampling random joint configurations and by evaluating the above expression

Tip: with px, py & pz lists of sampled TCP positions you can plot the workspace with following code fragment:

```
import pylab as pl
fig = pl.figure()
ax = pl.axes(projection="3d")
ax.scatter3D(px,py,pz,c=z,cmap='hsv') pl.show()
```

Requirement: have an entrypoint function:
def forward_kinematics(q: np.ndarray)